ShreyaRamesh-FinalProjectCheckIn

```
#Set default figure dimensions and formatting
knitr::opts_chunk$set(
  fig.width = 10,
  fig.height = 6,
  out.width = "100%",
  warning = FALSE,
  message = FALSE
)
```

Load packages and datasets

```
#Load packages
library(ggplot2)
library(ComplexHeatmap)

# Record R version and package versions
R.version.string
```

```
## [1] "R version 4.5.3 (2026-03-11)"
```

```
packageVersion("ggplot2")
```

```
## [1] '4.0.2'
```

```
packageVersion("ComplexHeatmap")
```

```
## [1] '2.26.1'
```

```
packageVersion("knitr")
```

```
## [1] '1.51'
```

```
#Load datasets
counts <- read.csv("counts.csv", row.name = 1)
metadata <- read.csv("meta_data.csv")

#Check & View
dim(counts)
```

```
## [1] 60660  1231
```

```r
dim(metadata)
```

```
## [1] 1231   98
```

Gene Selection and Summary Statistics

```r
#Select TP53 and BRCA1 as the project's genes
tp53 <- as.numeric(counts["ENSG00000141510.18", ])
brca1 <- as.numeric(counts["ENSG00000012048.23", ])

#Calculate summary statistics for both genes
gene_stats <- data.frame(
  Gene = c("TP53", "BRCA1"),
  Mean = c(mean(tp53), mean(brca1)),
  Median = c(median(tp53), median(brca1)),
  Standard_Deviation = c(sd(tp53), sd(brca1)),
  Minimum = c(min(tp53),min(brca1)),
  Maximum = c(max(tp53), max(brca1))
)

sample_type_summary <- as.data.frame(
  table(metadata$definition)
)

names(sample_type_summary) <- c("Sample_Type", "Sample_Count")

knitr::kable(
  sample_type_summary,
  caption = "Sample Counts by Sample Type"
)
```

Table 1: Sample Counts by Sample Type

Sample_Type	Sample_Count
Metastatic	7
Primary solid Tumor	1111
Solid Tissue Normal	113

```r
#Round numeric columns
gene_stats[, -1] <- round(gene_stats[, -1], 2)

#Display table
knitr::kable(
  gene_stats,
  caption = "Summary Statistics for TP53 and BRCA1 RNA-seq Counts"
)
```

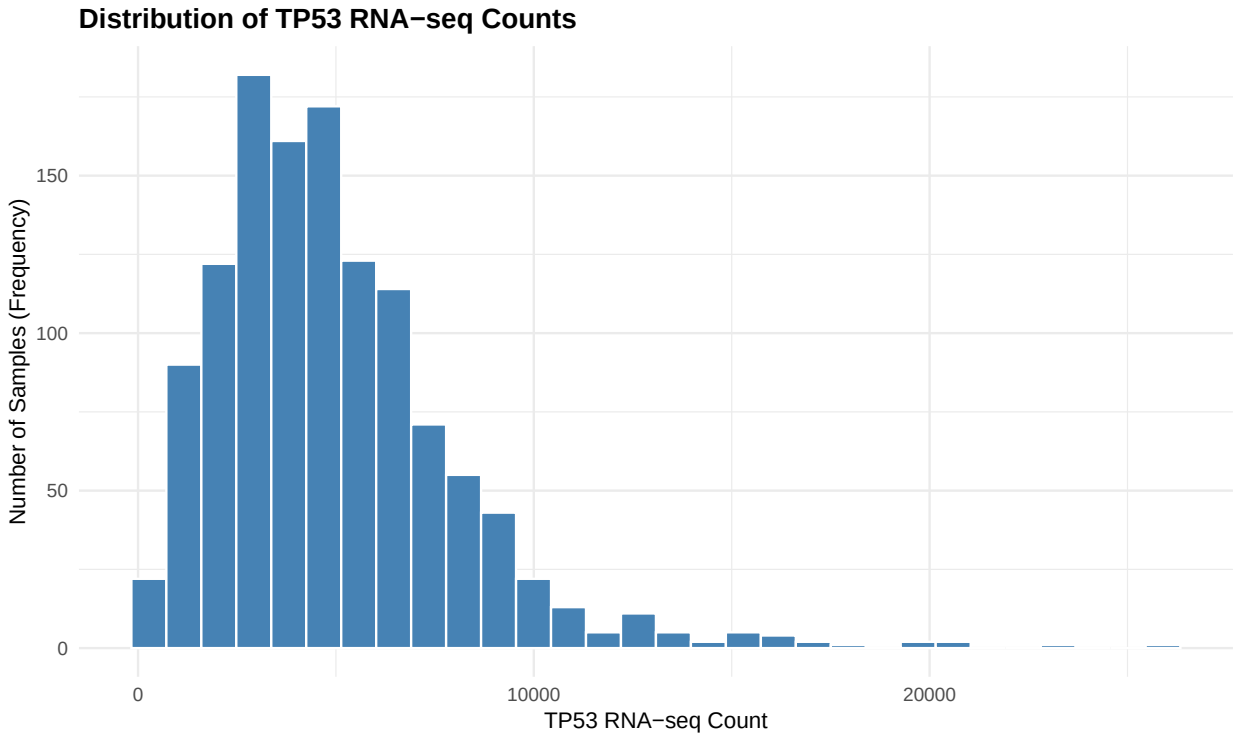
Table 2: Summary Statistics for TP53 and BRCA1 RNA-seq Counts

Gene	Mean	Median	Standard_Deviation	Minimum	Maximum
TP53	4935.74	4441	3036.71	274	25897
BRCA1	1304.21	919	1137.35	21	8050

Visualizations

```
#Create a dataframe
tp53_df <- data.frame(
  TP53_Count = tp53
)

#Create a histogram
ggplot(tp53_df, aes(x = TP53_Count)) +
  geom_histogram(
    bins = 30,
    fill = "steelblue",
    color = "white"
    ) +
  labs(
    title = "Distribution of TP53 RNA-seq Counts",
    x = "TP53 RNA-seq Count",
    y = "Number of Samples (Frequency)"
      ) +
  theme_minimal(base_size = 14) +
  theme(
    plot.title = element_text(face = "bold", size = 16),
    axis.title = element_text(size = 13),
    axis.text = element_text(size = 11)
  )
```

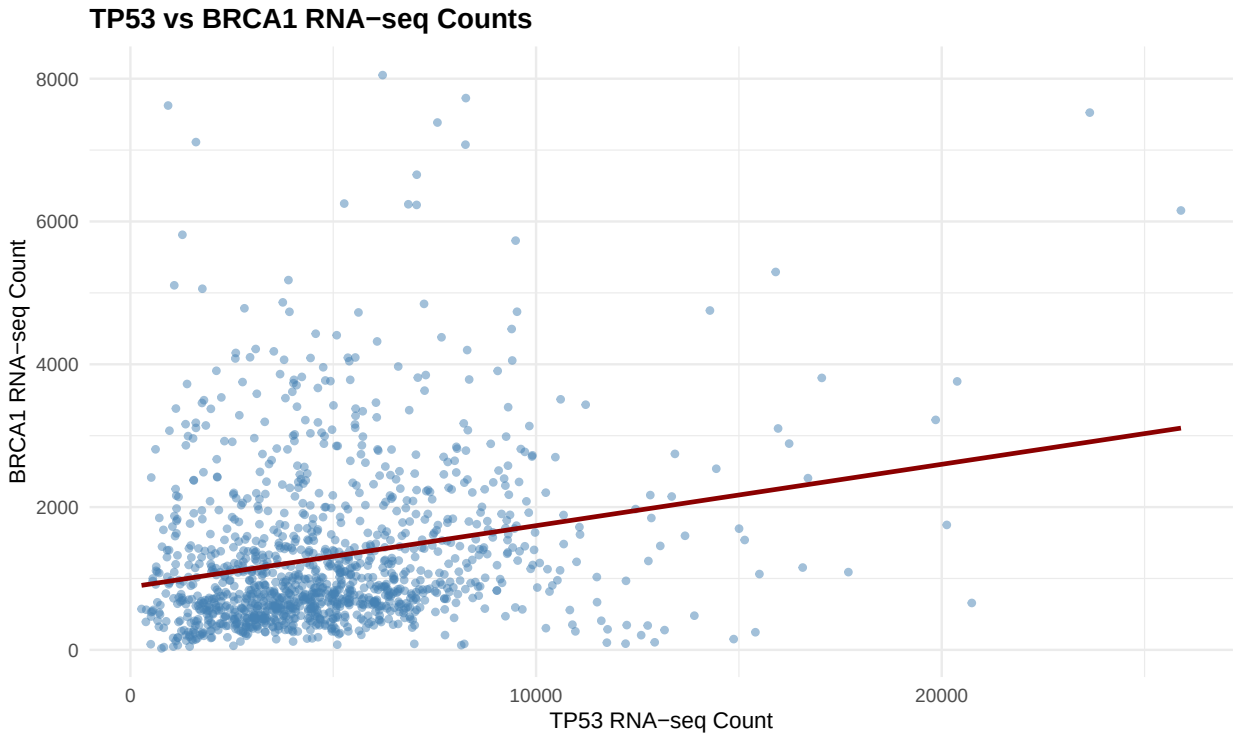



```
#Combine TP53 and BRCA1 counts
scatter_df <- data.frame(
  TP53 = tp53,
  BRCA1 = brca1
)
#Calculate Pearson correlation between TP53 and BRCA1
cor(tp53,brca1, method = "pearson")
```

```
## [1] 0.2295116
```

```
#Create a scatter plot
ggplot(scatter_df, aes(x = TP53, y = BRCA1)) +
  geom_point(
    alpha = 0.5,
    size = 1.5,
    color = "steelblue"
  ) +
  geom_smooth(
  method = "lm",
  se = FALSE,
  color = "darkred"
) +
  labs(
  title = "TP53 vs BRCA1 RNA-seq Counts",
  x = "TP53 RNA-seq Count",
  y = "BRCA1 RNA-seq Count"
) +
  theme_minimal(base_size = 14) +
  theme(
```

```
    plot.title = element_text(face = "bold", size = 16),
    axis.title = element_text(size = 13),
    axis.text = element_text(size = 11)
  )
```



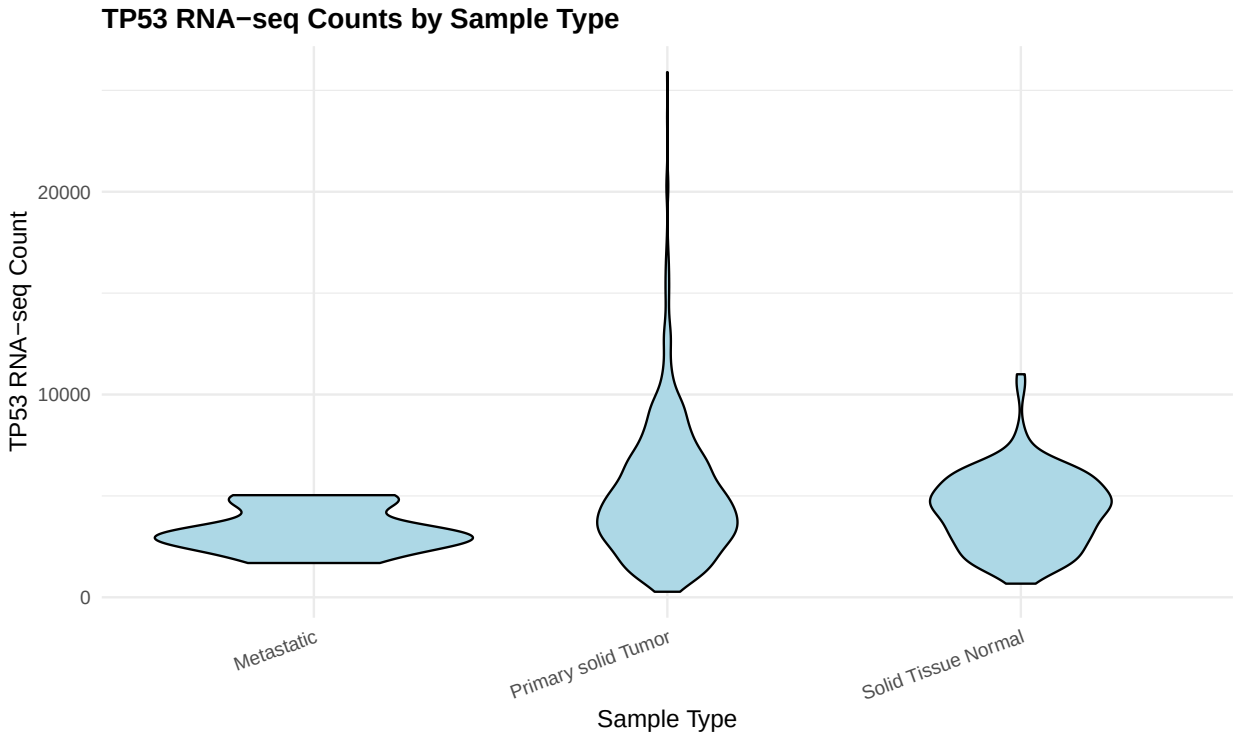

```
#Create a violin plot
tp53_samples <- data.frame(
  barcode = gsub("\\.", "-", colnames(counts)),
  TP53_Count = tp53
)

#Merge TP53 counts with the sample type metadata
violin_data <- merge(
  tp53_samples,
  metadata[, c("barcode", "definition")],
  by = "barcode"
)
dim(violin_data)
```

```
## [1] 1231    3
```

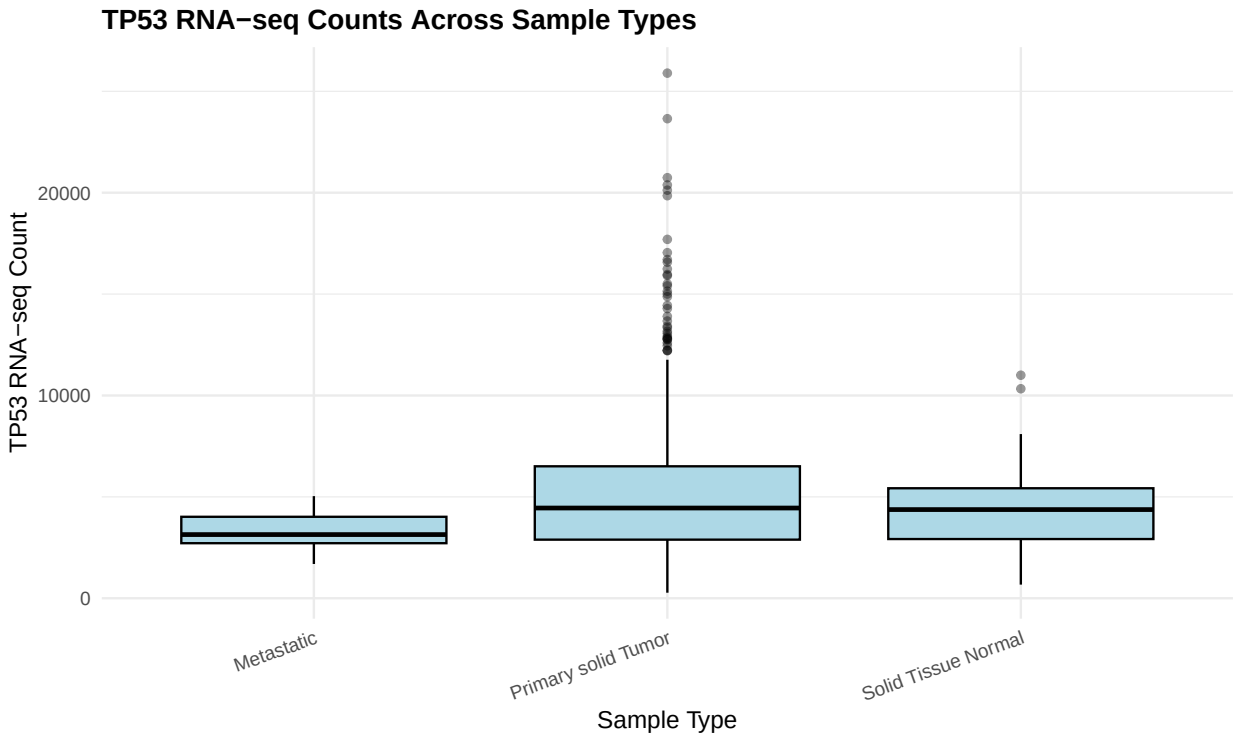
```
#Create a violin plot of TP53 counts by sample type
ggplot(violin_data, aes(x = definition, y = TP53_Count)) +
  geom_violin(
    fill = "lightblue",
    color = "black"
  ) +
  labs(
```

```
    title = "TP53 RNA-seq Counts by Sample Type",
    x = "Sample Type",
    y = "TP53 RNA-seq Count"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    plot.title = element_text(face = "bold", size = 16),
    axis.text.x = element_text(
      size = 11,
      angle = 20,
      hjust = 1
    )
  )
```

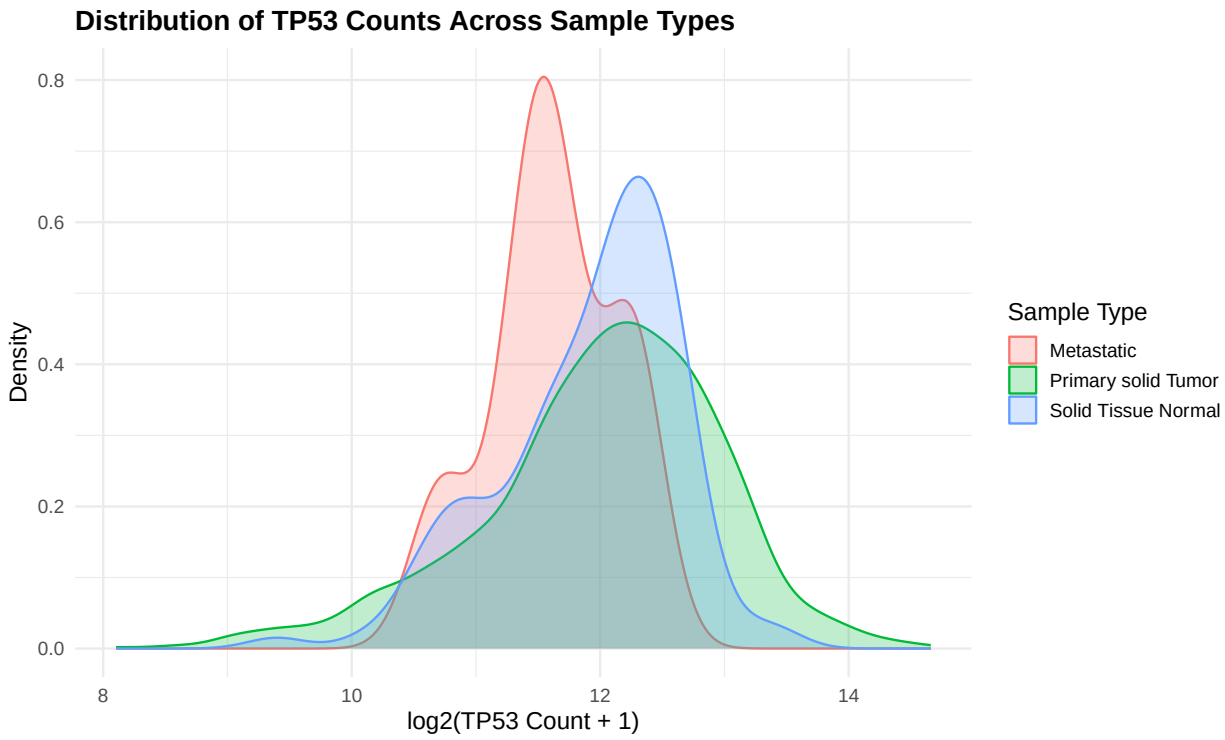



```
#Create a box plot
ggplot(violin_data, aes(x = definition, y = TP53_Count)) +
  geom_boxplot(
    fill = "lightblue",
    color = "black",
    outlier.alpha = 0.4
  ) +
  labs(
    title = "TP53 RNA-seq Counts Across Sample Types",
    x = "Sample Type",
    y = "TP53 RNA-seq Count"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    plot.title = element_text(face = "bold", size = 16),
```

```
    axis.text.x = element_text(angle = 20, hjust = 1)
  )
```




```
#Create a density plot
ggplot(
  violin_data,
  aes(
    x = log2(TP53_Count + 1),
    fill = definition,
    color = definition
  )
) +
  geom_density(alpha = 0.25) +
  labs(
    title = "Distribution of TP53 Counts Across Sample Types",
    x = "log2(TP53 Count + 1)",
    y = "Density",
    fill = "Sample Type",
    color = "Sample Type"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    plot.title = element_text(face = "bold", size = 16)
  )
```




```r
#Create a heatmap
#Select 10 cancer-related genes
selected_genes <- c(
  "ENSG00000141510.18",  # TP53
  "ENSG00000012048.23",  # BRCA1
  "ENSG00000139618.16",  # BRCA2
  "ENSG00000141736.14",  # ERBB2
  "ENSG00000091831.24",  # ESR1
  "ENSG00000146648.19",  # EGFR
  "ENSG00000171862.11",  # PTEN
  "ENSG00000136997.21",  # MYC
  "ENSG00000133703.13",  # KRAS
  "ENSG00000157764.14"   # BRAF
)
heatmap_data <- as.matrix(counts[selected_genes, ])
#Rename rows
rownames(heatmap_data) <- c(
   "TP53",
  "BRCA1",
  "BRCA2",
  "ERBB2",
  "ESR1",
  "EGFR",
  "PTEN",
  "MYC",
  "KRAS",
  "BRAF"
)
heatmap_log <- log2(heatmap_data + 1)
```

```r
#Add Annotation Bar
metadata_ordered <- metadata[
  match(
    gsub("\\.", "-", colnames(heatmap_log)),
    metadata$barcode
    ),
]
#Check avaliable metadata variables
names(metadata)
```

```
##  [1] "barcode"
##  [2] "patient"
##  [3] "sample"
##  [4] "shortLetterCode"
##  [5] "definition"
##  [6] "sample_submitter_id"
##  [7] "sample_type_id"
##  [8] "tumor_descriptor"
##  [9] "sample_id"
## [10] "sample_type"
## [11] "composition"
## [12] "days_to_collection"
## [13] "state"
## [14] "initial_weight"
## [15] "preservation_method"
## [16] "pathology_report_uuid"
## [17] "submitter_id"
## [18] "oct_embedded"
## [19] "specimen_type"
## [20] "is_ffpe"
## [21] "tissue_type"
## [22] "synchronous_malignancy"
## [23] "ajcc_pathologic_stage"
## [24] "days_to_diagnosis"
## [25] "laterality"
## [26] "treatments"
## [27] "tissue_or_organ_of_origin"
## [28] "age_at_diagnosis"
## [29] "primary_diagnosis"
## [30] "metastasis_at_diagnosis"
## [31] "prior_malignancy"
## [32] "year_of_diagnosis"
## [33] "prior_treatment"
## [34] "diagnosis_is_primary_disease"
## [35] "method_of_diagnosis"
## [36] "ajcc_staging_system_edition"
## [37] "ajcc_pathologic_t"
## [38] "morphology"
## [39] "ajcc_pathologic_n"
## [40] "ajcc_pathologic_m"
## [41] "classification_of_tumor"
## [42] "diagnosis_id"
## [43] "icd_10_code"
```

```
## [44] "site_of_resection_or_biopsy"
## [45] "sites_of_involvement"
## [46] "days_to_last_follow_up"
## [47] "follow_ups_disease_response"
## [48] "race"
## [49] "gender"
## [50] "ethnicity"
## [51] "vital_status"
## [52] "age_at_index"
## [53] "days_to_birth"
## [54] "demographic_id"
## [55] "age_is_obfuscated"
## [56] "country_of_residence_at_enrollment"
## [57] "bcr_patient_barcode"
## [58] "primary_site"
## [59] "project_id"
## [60] "disease_type"
## [61] "name"
## [62] "releasable"
## [63] "released"
## [64] "figo_stage"
## [65] "figo_staging_edition_year"
## [66] "days_to_death"
## [67] "tumor_of_origin"
## [68] "days_to_sample_procurement"
## [69] "last_known_disease_status"
## [70] "tumor_grade"
## [71] "progression_or_recurrence"
## [72] "alcohol_history"
## [73] "exposure_id"
## [74] "year_of_birth"
## [75] "paper_patient"
## [76] "paper_Tumor.Type"
## [77] "paper_Included_in_previous_marker_papers"
## [78] "paper_vital_status"
## [79] "paper_days_to_birth"
## [80] "paper_days_to_death"
## [81] "paper_days_to_last_followup"
## [82] "paper_age_at_initial_pathologic_diagnosis"
## [83] "paper_pathologic_stage"
## [84] "paper_Tumor_Grade"
## [85] "paper_BRCA_Pathology"
## [86] "paper_BRCA_Subtype_PAM50"
## [87] "paper_MSI_status"
## [88] "paper_HPV_Status"
## [89] "paper_tobacco_smoking_history"
## [90] "paper_CNV.Clusters"
## [91] "paper_Mutation.Clusters"
## [92] "paper_DNA.Methylation.Clusters"
## [93] "paper_mRNA.Clusters"
## [94] "paper_miRNA.Clusters"
## [95] "paper_lncRNA.Clusters"
## [96] "paper_Protein.Clusters"
## [97] "paper_PARADIGM.Clusters"
```

```
## [98] "paper_Pan.Gyn.Clusters"
```

```
#Examine sample type distribution
table(metadata$definition, useNA = "ifany")
```

```
##
##          Metastatic Primary solid Tumor Solid Tissue Normal
##                   7               1111                 113
```

```
#Check for variables related to stage or subtype
grep("stage|subtype", names(metadata),
     value = TRUE, ignore.case = TRUE)
```

```
## [1] "ajcc_pathologic_stage"    "figo_stage"
## [3] "paper_pathologic_stage"   "paper_BRCA_Subtype_PAM50"
```

```
#Check the number of missing values after matching
sum(is.na(metadata_ordered$ajcc_pathologic_stage))
```

```
## [1] 111
```

```
#Review the stage categories
table(metadata_ordered$ajcc_pathologic_stage,
      useNA = "ifany")
```

```
##
##    Stage 0    Stage I   Stage IA   Stage IB   Stage II  Stage IIA  Stage IIB
##          4         98         93          6          7        374        267
## Stage IIIA Stage IIIB Stage IIIC   Stage IV    Stage X       <NA>
##        155         27         62         20          7        111
```

```
# Verify sample metadata alignment
stopifnot(
  identical(
    gsub("\\.", "-", colnames(heatmap_log)),
    metadata_ordered$barcode
  )
)
sample_annotation <- HeatmapAnnotation(
  Sample_Type = metadata_ordered$definition,
  Pathologic_Stage = metadata_ordered$ajcc_pathologic_stage
)
#Generate the heatmap with the annotation
Heatmap(
  heatmap_log,
  name = "log2(Count + 1)",
  top_annotation = sample_annotation,
  cluster_rows = TRUE,
  cluster_columns = TRUE,
  show_column_names = FALSE,
  row_names_gp = grid::gpar(fontsize = 12),
```

```
  row_names_side = "left",
  column_title = "RNA-seq Count Patterns Across 10 Cancer-Related Genes",
  column_title_gp = grid::gpar(fontsize = 14, fontface = "bold"),
  heatmap_legend_param = list(
    title_gp = grid::gpar(fontsize = 11),
    labels_gp = grid::gpar(fontsize = 10)
  )
)
```

